CIRCULAR PERMUTATIONS OF FINITE FIELDS WITH PRIMITIVE ADJACENT PRODUCTS

KNOWLEDGE SENSE

ABSTRACT. Let $q > 7$ be a prime power and let $a \in \mathbb{F}_q$. Z.-W. Sun conjectured that the nonzero elements of $\mathbb{F}_q$ admit a circular permutation $(x_1, \dots, x_{q-1})$ such that every element $a + x_i x_{i+1}$, with $x_q = x_1$, is primitive. Luo and She proved this for all a when $q > 18{,}888{,}871$, and for all q when $a = 0$. We show that the conjecture holds for all pairs (q, a) except $(q, a) = (11, 2)$, where it fails: in the associated graph two vertices of degree two have the same two neighbours. For the finite range we give two explicit families of Hamilton cycles. For odd q a member of the first family exists as soon as $a+x$ is primitive for two nonsquares x whose ratio generates the group of nonzero squares; for even q a member of the second family exists as soon as $a + x$ is primitive for three consecutive powers x of a primitive element. These constructions handle all but 31 of the 659,460,173 pairs with $a \neq 0$ that are not covered by the criterion of Luo and She; thirty of the remaining pairs are settled by explicit circular permutations, and the last one is the exception.

1. INTRODUCTION

Let $\mathbb{F}_q$ be the finite field with q elements. An element of $\mathbb{F}_q$ is *primitive* if it generates the cyclic group $\mathbb{F}_q^*$. In 2013 Sun posed a series of conjectures on permutations and circular permutations of finite fields subject to arithmetic conditions on adjacent elements [5]. One of them is the following.

Conjecture 1.1 (Sun [5, Conjecture 3.8])**.** *Let $q > 7$ be a prime power and let $a \in \mathbb{F}_q$. There is a circular permutation $(x_1, \dots, x_{q-1})$ of $\mathbb{F}_q^*$ such that the $q - 1$ elements*

$$a + x_1x_2, \quad a + x_2x_3, \quad \dots, \quad a + x_{q-2}x_{q-1}, \quad a + x_{q-1}x_1$$

are all primitive.

In [5] the conjecture is dated 17 September 2013. The $q - 1$ values $a + x_i x_{i+1}$ need not be distinct. Following Luo and She [3], let $G_{q,a}$ be the simple graph with vertex set $\mathbb{F}_q^*$ in which distinct vertices x and y are adjacent if and only if $a + xy$ is primitive. Since $q - 1 \geq 7$, the circular permutations required in Conjecture 1.1 are exactly the Hamilton cycles of $G_{q,a}$.

Luo and She [3, Theorem 1.2] proved that $G_{q,a}$ is Hamiltonian for every prime power $q > 18{,}888{,}871$ and every $a \in \mathbb{F}_q$, and for every $q > 4$ when $a = 0$. For $a \neq 0$ they combine an estimate for the number of edges between two sets of vertices with the theorem of Chvátal and Erdős [1], and their numerical case analysis is

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modelled on that of Cohen, Oliveira e Silva and Trudgian [2]; for $a = 0$ they give an explicit construction in the exponents of a primitive element. We complete the classification.

Theorem 1.2. *Let $q > 7$ be a prime power and let $a \in \mathbb{F}_q$. The graph $G_{q,a}$ is Hamiltonian if and only if $(q, a) \neq (11, 2)$. Equivalently, there is a circular permutation of $\mathbb{F}_q^*$ in which a plus the product of any two adjacent elements is primitive if and only if $(q, a) \neq (11, 2)$.*

Corollary 1.3. *Conjecture 1.1 holds for every pair (q, a) with $(q, a) \neq (11, 2)$, and it fails for $(q, a) = (11, 2)$.*

The failure for $(11, 2)$ has a short proof (Proposition 4.1): the primitive roots modulo 11 are $2, 6, 7, 8$, and in $G_{11,2}$ the vertices 2 and 9 both have exactly the neighbours 3 and 8. For comparison, Sun's example for $q = 11$ [5, Remark 3.8] is the circular permutation $(1, 9, 2, 4, 5, 8, 10, 3, 6, 7)$, which uses $a = -1$.

For $a \neq 0$ the analytic criterion of Luo and She (Proposition 2.4) shows that $G_{q,a}$ is Hamiltonian whenever $\varphi(q-1) \geq 3 \cdot 2^{\omega(q-1)}\sqrt{q}+2$. We prove by a short argument that every $q \geq 25{,}000{,}001$ satisfies this inequality (Proposition 6.1); exactly 2,693 prime powers q with $8 \leq q \leq 25{,}000{,}000$ fail it, the largest being 18,888,871. For these orders we need the following explicit constructions.

Theorem 1.4. *Let $q > 7$ be a prime power, let g be a primitive element of $\mathbb{F}_q$ and let $a \in \mathbb{F}_q$.*

1. *If q is odd and there are integers j, k with $\gcd\big(j, \frac{q-1}{2}\big) = 1$ such that $a+g^{2k+1}$ and $a + g^{2k+2j+1}$ are primitive, then $G_{q,a}$ is Hamiltonian.*
2. *If q is even and there is an integer k such that $a+g^k$, $a+g^{k+1}$ and $a+g^{k+2}$ are primitive, then $G_{q,a}$ is Hamiltonian.*

The Hamilton cycles behind Theorem 1.4 are given explicitly in Section 5; along them the products of adjacent vertices take only two or three values. For these 2,693 orders and $a \neq 0$, a short search for k (and, rarely, j) settles all but 31 pairs (q, a); thirty of them are settled by explicit circular permutations listed in Appendix B, and the remaining one is $(11, 2)$.

Contribution. The analytic criterion for $a \neq 0$ and the construction for $a = 0$ are due to Luo and She [3]; for even $q - 1$ the construction appears earlier, in the language of unitary addition Cayley graphs, in [4, Theorem 11(i)]. For completeness we include short proofs of both; the reduction in Section 6 uses the criterion exactly in the form of Proposition 2.4. The new results of this paper are the exceptional pair, Theorem 1.4, and the treatment of all pairs (q, a) with $a \neq 0$ and $q \leq 18{,}888{,}871$ that are not covered by the criterion. To our knowledge, Theorem 1.2 has not appeared in the literature.

Dependencies. Besides elementary facts on finite fields and multiplicative characters, the proof uses the theorem of Chvátal and Erdős and a finite computation, described in Section 7 and Appendix A. Reductions of this kind, in which a character-sum estimate leaves finitely many fields to be checked by computer, are standard for problems on primitive elements; see, for example, [2], and [6] for another conjecture of Sun treated in this way.

Organization. Section 2 collects notation and the criterion of Luo and She. Section 3 treats $a = 0$ and Section 4 the exceptional pair. Section 5 proves Theorem 1.4. Section 6 reduces the problem to a finite range, Section 7 describes the computation, and Section 8 completes the proof. The appendices describe the data and list the explicit cycles.

2. Preliminaries

Throughout, $q > 7$ is a prime power and $n = q - 1$. For a positive integer m we write $\omega(m)$ for the number of distinct prime divisors of m, φ for Euler's function, μ for the Möbius function,

$$W(m) = 2^{\omega(m)}, \qquad \theta(m) = \frac{\varphi(m)}{m}.$$

Thus $W(m)$ is the number of squarefree divisors of m. If g is a primitive element of $\mathbb{F}_q$, then g^e is primitive if and only if $\gcd(e, n) = 1$, and $\mathbb{F}_q$ has exactly $\varphi(n)$ primitive elements. For a graph G we write $\alpha(G)$ for its independence number and $\kappa(G)$ for its vertex connectivity, with $\kappa(K_N) = N - 1$.

Lemma 2.1. *Let $\sigma\colon \mathbb{F} \to \mathbb{F}'$ be an isomorphism of finite fields with q elements. Then σ restricts to a graph isomorphism from $G_{\mathbb{F},a}$ onto $G_{\mathbb{F}',\sigma(a)}$ for every $a \in \mathbb{F}$. In particular, it suffices to prove Theorem 1.2 for one field of each order.*

Proof. A field isomorphism preserves sums, products and multiplicative orders, hence primitive elements and the adjacency relation. □

Theorem 2.2 (Chvátal and Erdős [1, Theorem 1])**.** *Let G be a graph with at least three vertices. If $\kappa(G) \geq \alpha(G)$, then G is Hamiltonian.*

We extend every multiplicative character χ of $\mathbb{F}_q$, including the principal character χ_0, by $\chi(0) = 0$. For $z \in \mathbb{F}_q$ the indicator function of the primitive elements is

$$\mathbf{1}_{\mathrm{prim}}(z) = \theta(n) \sum_{d \mid n} \frac{\mu(d)}{\varphi(d)} \sum_{\operatorname{ord}\chi = d} \chi(z), \tag{2.1}$$

where the inner sum runs over the $\varphi(d)$ characters of order d; see [3, (1)] and [2, Section 2]. Only squarefree d contribute.

For $a \in \mathbb{F}_q$ and $X, Y \subseteq \mathbb{F}_q^*$ put

$$E(X, Y) = \#\{(x, y) \in X \times Y : a + xy \text{ is primitive}\}.$$

If X and Y are disjoint, $E(X, Y)$ is the number of edges of $G_{q,a}$ between X and Y.

Lemma 2.3 (Luo and She [3, Lemma 2.2])**.** *Let $a \neq 0$ and $X, Y \subseteq \mathbb{F}_q^*$. Then*

$$\bigl|E(X, Y) - \theta(n)|X||Y|\bigr| \leq \theta(n)\, W(n) \sqrt{q|X||Y|}.$$

Proof. Let $\chi \neq \chi_0$. We first show that

$$\sum_{x \in \mathbb{F}_q} \Bigl|\sum_{y \in Y} \chi(a + xy)\Bigr|^2 = q|Y| - \Bigl|\sum_{y \in Y} \chi(y)\Bigr|^2. \tag{2.2}$$

Expanding the square, a diagonal pair $y = z$ contributes $\#\{x \in \mathbb{F}_q : a + xy \neq 0\} = q - 1$. Let $y \neq z$ be elements of Y. The fractional linear map $w(x) = (a + xy)/(a + xz)$

has determinant $a(y-z)\neq 0$, so it maps $\mathbb{F}_q\setminus\{-a/z\}$ bijectively onto $\mathbb{F}_q\setminus\{y/z\}$. The term with $x=-a/z$ vanishes because $\overline{\chi}(0)=0$, and $\sum_{w\in\mathbb{F}_q}\chi(w)=0$. Hence

$$\sum_{x\in\mathbb{F}_q}\chi(a+xy)\overline{\chi(a+xz)}=\sum_{w\neq y/z}\chi(w)=-\chi(y)\overline{\chi(z)},$$

and summing over all pairs gives $(q-1)|Y|-|\sum_y\chi(y)|^2+|Y|$, which is (2.2). By the Cauchy–Schwarz inequality and (2.2),

$$\Big|\sum_{x\in X}\sum_{y\in Y}\chi(a+xy)\Big|^2\leq|X|\sum_{x\in\mathbb{F}_q}\Big|\sum_{y\in Y}\chi(a+xy)\Big|^2\leq q|X||Y|.$$

For the principal character, $\sum_{x\in X}\sum_{y\in Y}\chi_0(a+xy)=|X||Y|-T_0$, where T_0 is the number of pairs $(x,y)\in X\times Y$ with $xy=-a$. Each x occurs in at most one such pair and so does each y, hence $0\leq T_0\leq\min(|X|,|Y|)\leq\sqrt{|X||Y|}$. Summing (2.1) over $X\times Y$ and using that there are $W(n)-1$ squarefree divisors $d>1$ of n, each with $\varphi(d)$ characters of order d, we obtain

$$\begin{aligned}\big|E(X,Y)-\theta(n)|X||Y|\big|&\leq\theta(n)\Big(\sqrt{|X||Y|}+(W(n)-1)\sqrt{q|X||Y|}\Big)\\&\leq\theta(n)W(n)\sqrt{q|X||Y|}.\end{aligned}$$

□

Proposition 2.4 (Luo and She [3, Lemmas 3.1, 3.2 and (4)])**.** *Let $a\neq 0$ and $G=G_{q,a}$. Then $\alpha(G)\leq 2W(n)\sqrt{q}+1$, and if G is not complete, then $\kappa(G)\geq\varphi(n)-W(n)\sqrt{q}-1$. Consequently, if*

$$\varphi(n)\geq 3W(n)\sqrt{q}+2, \tag{2.3}$$

then $G_{q,a}$ is Hamiltonian.

Proof. Let I be an independent set with $s=|I|\geq 2$, and split it into disjoint sets X and Y of sizes $\lfloor s/2\rfloor$ and $\lceil s/2\rceil$. Then $E(X,Y)=0$, and Lemma 2.3 gives $\sqrt{|X||Y|}\leq W(n)\sqrt{q}$. Since $|X||Y|=\lfloor s^2/4\rfloor\geq(s^2-1)/4$, we get $s\leq\sqrt{4W(n)^2q+1}\leq 2W(n)\sqrt{q}+1$.

For a vertex v, the map $y\mapsto a+vy$ is a bijection from $\mathbb{F}_q^*$ onto $\mathbb{F}_q\setminus\{a\}$, which contains at least $\varphi(n)-1$ primitive elements; excluding $y=v$ removes at most one more. Hence every vertex has degree at least $\varphi(n)-2$. Suppose that G is not complete, and let S be a set of $\kappa(G)$ vertices such that $G-S$ is disconnected; $S=\emptyset$ if G itself is disconnected. Let A be a smallest component of $G-S$ and B the union of the other components, so that $|A|\leq|B|$ and $E(A,B)=0$. Lemma 2.3 gives $|A|\leq\sqrt{|A||B|}\leq W(n)\sqrt{q}$. A vertex of A has all its neighbours in $(A\cup S)$ minus itself, so $\varphi(n)-2\leq|A|-1+|S|$, that is, $\kappa(G)\geq\varphi(n)-W(n)\sqrt{q}-1$.

If G is complete it is Hamiltonian, since $q-1\geq 3$. Otherwise (2.3) gives $\kappa(G)\geq 2W(n)\sqrt{q}+1\geq\alpha(G)$, and Theorem 2.2 applies. □

3. The zero offset

The following construction is due to Luo and She [3, Section 4]; for even n it is the Hamilton cycle of [4, Theorem 11(i)]. We include the short verification.

Proposition 3.1 (Luo and She; Sinha, Garg and Singh for even n)**.** *For every prime power $q>7$ the graph $G_{q,0}$ is Hamiltonian.*

Proof. Let g be a primitive element. Writing vertices as g^e with $e \in \mathbb{Z}/n\mathbb{Z}$, the product $g^e g^{e'}$ is primitive if and only if $e+e'$ is a unit modulo n. It suffices to give a cyclic ordering of $\mathbb{Z}/n\mathbb{Z}$ in which all cyclically adjacent sums are units.

If n is even, take $(0,\, n-1,\, 2,\, n-3,\, \ldots,\, n-2,\, 1)$, the concatenation of the pairs $(2j,\, n-1-2j)$ for $0 \le j \le n/2-1$. Adjacent sums are $n-1$ inside a pair, $n+1$ between consecutive pairs, and $1+0=1$ at the closing step; all of them are ± 1 modulo n.

If n is odd, then q is even and $n = 2h+1$ with $h \ge 3$. Take

$$(n-1,\ 0,\ n-2,\ 1,\ h) \quad \text{followed by the pairs} \quad (h-j,\ h+j), \qquad 1 \le j \le h-2.$$

The pairs contain $2, \ldots, h-1$ and $h+1, \ldots, n-3$, so this is a permutation of $\mathbb{Z}/n\mathbb{Z}$; for $n=7$ it is $(6,0,5,1,3,2,4)$. The adjacent sums are $n-1$, $n-2$, $n-1$, $h+1$ in the first block, $2h-1 = n-2$ from h to $h-1$ and between consecutive pairs, $2h = n-1$ inside a pair, and $(n-3)+(n-1) \equiv n-4$ at the closing step. Since n is odd, $n-1$, $n-2$ and $n-4$ are units modulo n, and so is $h+1$, because $2(h+1) = n+1$. □

Remark 3.2. The map $g^e \mapsto e$ identifies $G_{q,0}$ with the unitary addition Cayley graph of $\mathbb{Z}/n\mathbb{Z}$, in which distinct e, e' are adjacent when $e+e'$ is a unit [4]. The odd case of [4, Theorem 11] assumes $3 \nmid n$, which fails for instance for $n = 15$ and $n = 63$; the construction above covers every odd $n \ge 7$.

4. The exceptional pair

Proposition 4.1. *The graph $G_{11,2}$ is not Hamiltonian.*

Proof. The primitive elements of $\mathbb{F}_{11}$ are $2, 6, 7, 8$. Since $xy \neq 0$ for $x, y \in \mathbb{F}_{11}^*$, the vertices $x \neq y$ are adjacent if and only if $xy \in \{4,5,6\}$. Solving $2y \in \{4,5,6\}$ gives $y \in \{2,8,3\}$, and solving $9y \in \{4,5,6\}$ gives $y \in \{9,3,8\}$. Excluding loops, both 2 and 9 have exactly the neighbours 3 and 8. A Hamilton cycle must use both edges at every vertex of degree two, so it contains the cycle $2,3,9,8,2$. This cycle has four vertices, while $G_{11,2}$ has ten, a contradiction. □

5. Two families of Hamilton cycles

Lemma 5.1. *Let q be odd, $m = (q-1)/2$, let h generate the subgroup of nonzero squares and let x be a nonsquare. Put*

$$y_{2i} = h^i, \qquad y_{2i+1} = xh^{-i} \qquad (0 \le i \le m-1).$$

Then $(y_0, y_1, \ldots, y_{q-2})$ is a circular permutation of $\mathbb{F}_q^*$, and every product of two cyclically adjacent terms equals x or xh. In particular, if $a+x$ and $a+xh$ are primitive, then $(y_0, \ldots, y_{q-2})$ is a Hamilton cycle of $G_{q,a}$.

Proof. The terms y_{2i} run through the m squares and the terms y_{2i+1} through the coset $x\langle h\rangle$, which consists of the m nonsquares. Furthermore $y_{2i}y_{2i+1} = x$, $y_{2i+1}y_{2i+2} = xh$ for $0 \le i \le m-2$, and $y_{q-2}y_0 = xh^{-(m-1)} = xh$ because $h^m = 1$. □

Lemma 5.2. *Let q be even, let g be primitive and $x \in \mathbb{F}_q^*$, and let r be the square root of x. If $a+x$ and $a+xg^2$ are primitive, then $G_{q,a}$ has a Hamilton path from r to gr along which every product of adjacent vertices equals x or xg^2. If moreover $a+xg$ is primitive, then $G_{q,a}$ is Hamiltonian.*

Proof. Since squaring is a bijection of $\mathbb{F}_q$, the element r is well defined. Consider the involutions $F(y) = x/y$ and $J(y) = xg^2/y$ of $\mathbb{F}_q^*$, and let H be the graph on $\mathbb{F}_q^*$ with edges $\{y, F(y)\}$ for $F(y) \neq y$ and $\{y, J(y)\}$ for $J(y) \neq y$. Since n is odd and $n > 2$, we have $g^2 \neq 1$, so $F(y) \neq J(y)$ for all y. The unique fixed point of F is r and that of J is gr, and $r \neq gr$. Hence r and gr have degree one in H and every other vertex has degree two. The products along edges of H are x and xg^2, so under the hypothesis every edge of H is an edge of $G_{q,a}$.

Each component of H is invariant under F and J, hence under JF, which is multiplication by g^2. As n is odd, g^2 generates $\mathbb{F}_q^*$, so JF acts transitively on the vertices and H is connected. A connected graph of maximum degree two with exactly two vertices of degree one is a path; thus H is a Hamilton path from r to gr. Finally $r \cdot gr = xg$. If $a + xg$ is also primitive, then r and gr are adjacent in $G_{q,a}$, and since $q - 1 \geq 3$ they are not adjacent on the path, so adding this edge closes a Hamilton cycle. □

Proof of Theorem 1.4. (1) Let $m = (q-1)/2$. The order of g^{2j} is $2m/\gcd(2j, 2m) = m/\gcd(j, m) = m$, so $h = g^{2j}$ generates the squares, and $x = g^{2k+1}$ is a nonsquare with $xh = g^{2k+2j+1}$. Apply Lemma 5.1. (2) Apply Lemma 5.2 with $x = g^k$. □

6. Reduction to a finite range

Luo and She showed that (2.3) holds for every $q > 18{,}888{,}871$ by a case analysis according to $\omega(q-1)$ [3, Section 3]. To make the finite range independently checkable we use the following weaker statement, whose proof reduces to two explicit integer inequalities. Let P_r denote the product of the first r primes $p_1 < p_2 < \cdots < p_r$.

Proposition 6.1. *Every prime power $q \geq 25{,}000{,}001$ satisfies* (2.3).

Proof. Let $n = q - 1 = \prod_{i=1}^{r} \ell_i^{a_i}$ with primes $\ell_1 < \cdots < \ell_r$, so that $\ell_i \geq p_i$. Since $t \mapsto (t-1)/\sqrt{t}$ is increasing for $t \geq 1$,

$$\frac{\varphi(n)}{n} = \prod_{i=1}^{r}\Big(1 - \frac{1}{\ell_i}\Big) \geq \frac{\varphi(P_r)}{P_r}, \qquad \frac{\varphi(n)}{2^r\sqrt{n}} = \prod_{i=1}^{r} \ell_i^{(a_i-1)/2}\,\frac{\ell_i - 1}{2\sqrt{\ell_i}} \geq \prod_{i=1}^{r} \frac{p_i - 1}{2\sqrt{p_i}}. \tag{6.1}$$

Suppose first that $r \leq 8$. Then $W(n) \leq 256$ and $\varphi(n) \geq cn$ with $c = \varphi(P_8)/P_8$, where $P_8 = 9{,}699{,}690$ and $\varphi(P_8) = 1{,}658{,}880$. It suffices to show that $(cn - 2)/\sqrt{n+1} \geq 768$. The function $t \mapsto (ct - 2)/\sqrt{t+1}$ is increasing for $t > 0$, since its derivative is $(ct + 2c + 2)/(2(t+1)^{3/2})$. For $M = 25{,}000{,}000 \leq n$ we have $\varphi(P_8)M - 2P_8 = 41{,}471{,}980{,}600{,}620 > 0$ and

$$(\varphi(P_8)M - 2P_8)^2 - 9 \cdot 256^2 (M+1) P_8^2 = 332{,}600{,}294{,}066{,}556{,}440{,}798{,}298{,}000 > 0,$$

which is the required inequality at $t = M$ multiplied by P_8 and squared.

Suppose now that $r \geq 9$. Then $n \geq P_9 = 223{,}092{,}870$, $\varphi(P_9) = 36{,}495{,}360$, and

$$\varphi(P_9)^2 - 16 \cdot 512^2\, P_9 = 396{,}191{,}984{,}517{,}120 > 0,$$

so the product in (6.1) over $i \leq 9$ exceeds 4. Each further factor exceeds 1, because $(p-1)^2 > 4p$ for $p \geq 29$. Hence $\varphi(n) > 4W(n)\sqrt{n}$. Since $\sqrt{n+1} \leq \frac{5}{4}\sqrt{n}$ and $W(n)\sqrt{n}/4 > 2$,

$$\varphi(n) > \tfrac{15}{4} W(n)\sqrt{n} + \tfrac{1}{4} W(n)\sqrt{n} > 3W(n)\sqrt{n+1} + 2. \qquad \square$$

For $8 \le q \le 25{,}000{,}000$ we have $\varphi(q-1) \ge 2$, so (2.3) fails exactly when

$$(\varphi(q-1)-2)^2 < 9 \cdot 4^{\omega(q-1)}\, q. \tag{6.2}$$

There are 1,566,722 prime powers q with $8 \le q \le 25{,}000{,}000$, and exactly 2,693 of them satisfy (6.2): 2,617 primes and 76 proper prime powers. We call them the *residual orders*. Their sum is 659,462,866, and seven of them are powers of two, namely 2^d for $d \in \{3, 4, 5, 6, 8, 10, 12\}$. Two independently written sieves produce the same list; Table 6.1 shows its distribution.

TABLE 6.1. Residual orders $q \le 25{,}000{,}000$, that is, prime powers satisfying (6.2), by the value of $\omega(q-1)$.

$\omega(q-1)$	residual orders	largest residual order
1	4	32
2	66	1,297
3	332	8,101
4	821	44,101
5	913	204,751
6	453	960,961
7	99	4,084,081
8	5	18,888,871
total	2,693	

The largest residual order is 18,888,871, so the threshold in [3, Theorem 1.2] is exactly the largest prime power that fails (2.3).

Corollary 6.2. *Let $q > 7$ and $a \in \mathbb{F}_q$. If $G_{q,a}$ is not Hamiltonian, then $a \ne 0$ and q is a residual order.*

Proof. Combine Propositions 3.1, 2.4 and 6.1. □

7. The finite computation

7.1. **Field models.** For each residual order $q = p^d$ we fix a monic polynomial $f \in \mathbb{F}_p[T]$ of degree d (with $f = T$ when $d = 1$) and an element g of $R = \mathbb{F}_p[T]/(f)$. A residue class $\sum_{i<d} c_i T^i$ with $0 \le c_i < p$ is encoded by the integer $\sum_{i<d} c_i p^i$, so that R is identified with $\{0, \dots, q-1\}$; for $d = 1$ this is the usual model $\mathbb{Z}/p\mathbb{Z}$. The polynomials for $q \in \{9, 16, 25\}$ are listed in Appendix B.

Lemma 7.1. *If the powers g^e, $0 \le e \le q-2$, are pairwise distinct and nonzero and $g^{q-1} = 1$, then R is a field, f is irreducible, g is primitive, and g^e is primitive if and only if* $\gcd(e, q-1) = 1$.

Proof. The $q-1$ powers exhaust $R \setminus \{0\}$, and g is a unit since $g \cdot g^{q-2} = 1$. Hence every nonzero element of R is a unit, so R is a field with q elements and f is irreducible. The group R^* is cyclic of order $q-1$ generated by g. □

The checking program verifies that p is prime and $q = p^d$, checks the hypothesis of Lemma 7.1 by computing the whole orbit of g, and marks g^e as primitive exactly when $\gcd(e, q-1) = 1$. The element g is the primitive element with the least encoding, which the program also checks.

7.2. **Witness search.** Fix a residual order q and an offset $a \in \mathbb{F}_q^*$. If q is odd, let $m = (q-1)/2$ and let $j_1 < j_2 < \dots$ be the smallest integers j with $1 \le j \le \min(m-1, 64)$ and $\gcd(j, m) = 1$, at most eight of them, so that $j_1 = 1$. In stage s the program tests, for $k = 0, 1, \dots, L_s - 1$ in increasing order, whether $a + g^{2k+1}$ and $a + g^{2k+2j_s+1}$ are both primitive; this is the condition of Theorem 1.4(1). If q is even, there is a single stage, which tests, for $k = 0, 1, \dots, L_1 - 1$, whether $a + g^k$, $a + g^{k+1}$ and $a + g^{k+2}$ are all primitive, the condition of Theorem 1.4(2). The stage lengths L_s are input values recorded in the certificate of each field; they satisfy $L_s \le \min(1024, m)$ for odd q and $L_1 \le \min(1024, q-1)$ for even q. Stages after the first are applied only to offsets not settled by an earlier stage.

The certificate of a field records the model (p, d, f, g), the stage lengths and the complete sorted list of offsets $a \in \mathbb{F}_q^*$ for which no stage succeeds. The program reruns the search with exactly these stage lengths for every $a \in \mathbb{F}_q^*$ and accepts the certificate only if the list of unsettled offsets agrees exactly with the recorded one. A successful test proves that $G_{q,a}$ is Hamiltonian by Theorem 1.4. If no stage succeeds, nothing is concluded from the search; such a pair is treated separately, either by an explicit Hamilton cycle that is checked directly, as described below, or, for $(11, 2)$, by Proposition 4.1.

7.3. **Results.** Since the residual orders sum to 659,462,866, there are 659,460,173 pairs (q, a) with q a residual order and $a \neq 0$. Table 7.1 summarizes the outcome of the search with the recorded stage lengths. Additional stages were needed in 16 fields, the largest being $q = 211$; in total they required 1,302 further tests and settled 71 offsets. The 31 offsets left by the search are listed in Table 7.2. For thirty of them Appendix B gives an explicit circular permutation of $\mathbb{F}_q^*$; the remaining one is $(11, 2)$. A second, independently written program checks these thirty cycles, using its own polynomial arithmetic and computing multiplicative orders directly: each cycle is a permutation of $\mathbb{F}_q^*$, and $a + xy$ is primitive for every pair of cyclically adjacent entries, including the closing pair. The same program checks the two neighbourhoods used in Proposition 4.1.

TABLE 7.1. Pairs (q, a) with q a residual order and $a \neq 0$.

residual orders	2,693
pairs (q, a) with $a \in \mathbb{F}_q^*$	659,460,173
settled by the first stage of the search	659,460,071
settled by additional stages	71
settled by an explicit cycle (Appendix B)	30
not Hamiltonian (Proposition 4.1)	1

Proposition 7.2. *Let q be a residual order and $a \in \mathbb{F}_q^*$. Then $G_{q,a}$ is Hamiltonian unless $(q, a) = (11, 2)$.*

Proof. By Lemma 7.1 each model is a field of order q, and by Lemma 2.1 it suffices to treat these models. Offsets settled by the search are covered by Theorem 1.4, and the thirty pairs of Table 7.2 other than $(11, 2)$ by the cycles of Appendix B. □

TABLE 7.2. Offsets not settled by the witness search, in the encoding of Section 7.

q	model	offsets a
9	$\mathbb{F}_3[T]/(T^2+1)$	$4,5,7,8$
11	$\mathbb{Z}/11\mathbb{Z}$	$2,3,4,8$
13	$\mathbb{Z}/13\mathbb{Z}$	$1,2,3,5,6,7,8,10,11,12$
16	$\mathbb{F}_2[T]/(T^4+T+1)$	$7,9,11$
25	$\mathbb{F}_5[T]/(T^2+2)$	$1,2,3,4,5,10,15,20$
31	$\mathbb{Z}/31\mathbb{Z}$	3
43	$\mathbb{Z}/43\mathbb{Z}$	5

8. Proof of the main theorem

Proof of Theorem 1.2. By Proposition 4.1, $G_{11,2}$ is not Hamiltonian. Conversely, let $(q,a) \neq (11,2)$. If $a = 0$, Proposition 3.1 applies. If $a \neq 0$ and q is not a residual order, then q satisfies (2.3): for $q \geq 25{,}000{,}001$ by Proposition 6.1, and for $q \leq 25{,}000{,}000$ by the definition of residual orders. Proposition 2.4 then applies. If q is a residual order, Proposition 7.2 applies. □

Proof of Corollary 1.3. The circular permutations in Conjecture 1.1 are the Hamilton cycles of $G_{q,a}$. □

Remark 8.1. In Theorem 1.4(1) only the residues of j and k modulo $m = (q-1)/2$ matter, and replacing (j,k) by $(-j,k+j)$ interchanges $a+g^{2k+1}$ and $a+g^{2k+2j+1}$. Hence, when k runs over all residues modulo m, the values j and $-j$ give the same condition. For $q = 13$ we have $m = 6$, and every admissible j is congruent to ± 1 modulo 6, so only $j = 1$ is relevant. As k runs over all residues modulo 6, the condition holds for only two of the twelve nonzero offsets; in particular, the second stage $j_2 = 5$ settles no further offset for $q = 13$.

Appendix A. Data availability

The list of residual orders, the field certificates, the thirty explicit cycles and the programs that check them are contained in the accompanying verification archive, together with a README file that describes how to reproduce the computations. The programs use only the Python standard library and a C++17 compiler. The list of residual orders records, for each residual order q, the characteristic, the degree, $\varphi(q-1)$, $\omega(q-1)$ and $W(q-1)$.

Appendix B. Explicit cycles

For $q \in \{11,13,31,43\}$ we use $\mathbb{Z}/q\mathbb{Z}$, and for $q \in \{9,16,25\}$ the models

$$\mathbb{F}_9 = \mathbb{F}_3[T]/(T^2+1), \qquad \mathbb{F}_{16} = \mathbb{F}_2[T]/(T^4+T+1), \qquad \mathbb{F}_{25} = \mathbb{F}_5[T]/(T^2+2).$$

An integer $x = \sum_i c_i p^i$ with $0 \leq c_i < p$ denotes the element $\sum_i c_i T^i$; for example, 7 denotes $1+2T$ in $\mathbb{F}_9$ and $1+T+T^2$ in $\mathbb{F}_{16}$. In these models the primitive elements are $4,5,7,8$ in $\mathbb{F}_9$, $2,3,4,5,9,11,13,14$ in $\mathbb{F}_{16}$, and $6,9,12,13,17,18,21,24$ in $\mathbb{F}_{25}$. For each pair (q,a) we list a circular permutation $(x_1,\dots,x_{q-1})$ of $\mathbb{F}_q^*$ such that $a + x_i x_{i+1}$ is primitive for all i, with $x_q = x_1$.

$(q, a) = (9, 4)$: (1, 3, 7, 8, 2, 6, 5, 4)

$(q, a) = (9, 5)$: (1, 3, 4, 7, 2, 6, 8, 5)

$(q, a) = (9, 7)$: (1, 6, 4, 5, 2, 3, 8, 7)

$(q, a) = (9, 8)$: (1, 6, 7, 4, 2, 3, 5, 8)

$(q, a) = (11, 3)$: (1, 4, 9, 3, 7, 10, 6, 8, 2, 5)

$(q, a) = (11, 4)$: (1, 2, 7, 6, 4, 5, 3, 8, 10, 9)

$(q, a) = (11, 8)$: (1, 5, 4, 8, 2, 10, 6, 7, 3, 9)

$(q, a) = (13, 1)$: (1, 6, 3, 12, 7, 2, 5, 9, 4, 8, 11, 10)

$(q, a) = (13, 2)$: (1, 4, 11, 2, 9, 12, 8, 6, 3, 10, 7, 5)

$(q, a) = (13, 3)$: (1, 4, 2, 8, 7, 3, 10, 6, 5, 11, 9, 12)

$(q, a) = (13, 5)$: (1, 6, 9, 3, 2, 5, 8, 4, 7, 12, 11, 10)

$(q, a) = (13, 6)$: (1, 5, 8, 12, 4, 10, 7, 2, 11, 6, 3, 9)

$(q, a) = (13, 7)$: (1, 8, 7, 11, 9, 10, 3, 4, 2, 6, 5, 12)

$(q, a) = (13, 8)$: (1, 3, 4, 5, 11, 7, 6, 2, 8, 9, 10, 12)

$(q, a) = (13, 10)$: (1, 10, 7, 5, 2, 11, 4, 12, 8, 6, 3, 9)

$(q, a) = (13, 11)$: (1, 4, 2, 11, 9, 12, 5, 6, 10, 3, 7, 8)

$(q, a) = (13, 12)$: (1, 8, 9, 10, 2, 4, 3, 11, 5, 12, 6, 7)

$(q, a) = (16, 7)$: (1, 2, 5, 8, 13, 12, 14, 6, 3, 7, 9, 4, 11, 15, 10)

$(q, a) = (16, 9)$: (1, 2, 5, 6, 4, 9, 14, 7, 11, 12, 8, 3, 10, 15, 13)

$(q, a) = (16, 11)$: (1, 5, 3, 7, 2, 11, 13, 6, 8, 14, 10, 4, 9, 12, 15)

$(q, a) = (25, 1)$: (1, 11, 3, 16, 6, 10, 9, 14, 24, 15, 2, 19, 21, 17, 8, 12, 23, 5, 13, 22, 4, 7, 18, 20)

$(q, a) = (25, 2)$: (1, 7, 15, 24, 8, 21, 4, 6, 22, 9, 10, 23, 2, 12, 5, 16, 17, 19, 13, 3, 14, 18, 20, 11)

$(q, a) = (25, 3)$: (1, 6, 8, 21, 10, 7, 23, 24, 4, 9, 15, 22, 19, 2, 13, 20, 16, 12, 3, 11, 17, 5, 18, 14)

$(q, a) = (25, 4)$: (1, 7, 5, 18, 8, 17, 4, 12, 22, 13, 20, 23, 3, 6, 14, 15, 9, 16, 2, 24, 10, 21, 11, 19)

$(q, a) = (25, 5)$: (1, 4, 14, 3, 7, 8, 18, 9, 2, 19, 11, 16, 23, 22, 12, 21, 5, 10, 20, 15, 6, 17, 24, 13)

$(q, a) = (25, 10)$: (1, 8, 14, 3, 11, 23, 7, 2, 16, 4, 19, 22, 10, 6, 5, 21, 17, 15, 9, 13, 12, 18, 20, 24)

$(q, a) = (25, 15)$: (1, 9, 5, 11, 7, 19, 23, 2, 18, 15, 10, 17, 12, 13, 3, 16, 22, 8, 14, 4, 24, 21, 20, 6)

$(q, a) = (25, 20)$: (1, 4, 7, 17, 6, 11, 2, 8, 12, 5, 10, 14, 23, 3, 16, 19, 22, 20, 13, 21, 9, 15, 24, 18)

$(q, a) = (31, 3)$: (1, 9, 8, 14, 5, 10, 7, 2, 25, 19, 4, 20, 16, 18, 28, 24, 3, 15, 13, 11, 27, 12, 6, 29, 22, 26, 17, 23, 30, 21)

$(q, a) = (43, 5)$: (1, 14, 23, 5, 10, 20, 33, 38, 3, 24, 31, 6, 37, 12, 19, 40, 15, 41, 29, 21, 13, 11, 17, 32, 2, 25, 4, 16, 9, 36, 35, 26, 34, 8, 7, 39, 27, 18, 30, 22, 42, 28)

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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